

# VA-ATACOM Supplementary Material: The DT-margin Heuristic Baseline

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Table 1: DT-margin ablation on Goal (5 seeds).

| Config                  | $\alpha$   | $h$      | Mean $\bar{C}$ | GO/5       |
|-------------------------|------------|----------|----------------|------------|
| Velocity-adaptive only  | 0.3        | 0        | 2.90           | 5/5        |
| BRT-only                | 0.0        | 3        | 2.35           | 4/5        |
| <b>DT-margin (full)</b> | <b>0.3</b> | <b>3</b> | <b>0.96</b>    | <b>5/5</b> |

## The DT-margin Heuristic Baseline

VA-ATACOM’s velocity augmentation has a heuristic predecessor we call the **DT-margin** filter — the “DT-margin (heuristic)” row of Table 1 in the main paper. It inflates the keepout radius linearly with speed,  $r_{\text{eff}}(\mathbf{v}) = r_{\text{base}}(1 + \alpha \|\mathbf{v}\|)$ , projects out the inward radial component, and adds the same sim-BRT reward shaping. As shown in the main paper’s design study, this linear margin is the *first-order* approximation of VA-ATACOM’s quadratic braking constraint; under fair hazard-radius matching it reaches 13/15 GO — two Push seeds short of VA-ATACOM’s 15/15, illustrating that the linearisation almost captures the right behaviour but leaves a residual gap that only the exact quadratic constraint closes. We characterise it here for completeness — its two tunable knobs ( $\alpha$  and the BRT horizon  $h$ ) and their interaction with the control rate. These findings motivated, and are superseded by, the parameter-free velocity-augmented manifold of the main paper.

## Ablation of the two components

Table 1 (Fig. 1) shows both components contribute: velocity-adaptive margin alone reaches 5/5 GO at higher variance (2.90); BRT alone 4/5 GO (one M1 collapse at  $\bar{C}=7.98$ ); the full heuristic 0.96 cost / 100% GO.

## $\Delta t$ sensitivity

Table 2 corroborates the M1 chord-excursion premise of the main paper’s failure-mode analysis (and hence Prop. 4): continuous-time ATACOM achieves zero cost at  $\Delta t \leq 0.02\text{s}$  but fails at  $\Delta t \geq 0.05\text{s}$ , the phase transition matching the M1 threshold  $\Delta t \|u\| \gtrsim \sqrt{d_{\text{safe}}} \cdot r$ . The DT-margin heuristic tracks the safe regime at  $\Delta t \in \{0.10, 0.20\}\text{s}$  with  $h=3$ . At  $\Delta t=0.05\text{s}$  the picture is nuanced: the horizon rule prescribes  $h^*=4$ , but a 5-seed sweep finds  $h=3$  at 3/5 GO (bimodal),  $h=4$  at 0/5, and  $h=5$  at 3/5 (one seed at 37.05). We attribute

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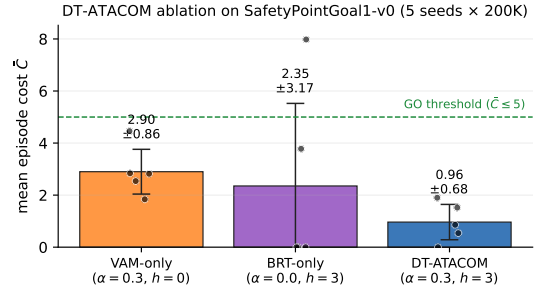


Figure 1: DT-margin ablation: both components are necessary. Mean episode cost  $\bar{C}$  on SafetyPointGoal1-v0 (5 seeds  $\times$  200K env steps), bars are  $\pm 1$  std. VAM-only ( $\alpha=0.3, h=0$ ) addresses M1 chord excursion only; BRT-only ( $\alpha=0, h=3$ ) anticipates M2 drift but one seed collapses via M1; DT-margin ( $\alpha=0.3, h=3$ ) combines both for lowest mean (0.96) and 5/5 GO.

Table 2: DT-margin cost vs. time step ( $\bar{C} \pm \sigma$ , 5 seeds, 50K steps).

| $\Delta t$ | ATACOM | DT-margin ( $h=3$ )      | Note         |
|------------|--------|--------------------------|--------------|
| 0.01s      | 0.00   | 3.02                     | ATACOM OK    |
| 0.02s      | 0.00   | 0.00                     | ATACOM OK    |
| 0.05s      | 39.95  | 4.96 $\pm$ 6.08 (3/5 GO) | Basin regime |
| 0.10s      | 13.68  | <b>2.60</b>              | DT-margin OK |
| 0.20s      | —      | 1.33 $\pm$ 1.50          | $h \gg h^*$  |

this to a *BRT-shaping basin*: near the penetration horizon the lookahead reports unavoidable penetration along multiple directions every step, integrating spurious penalty into the policy gradient and trapping training.

## $\alpha/h$ sensitivity

Sweeping  $\alpha$  and  $h$  on Goal (5 seeds  $\times$  200K):

Only configurations at least one step above the minimum on *both* axes ( $\alpha \geq 0.1, h \geq 2$ ) reach 5/5 GO; setting  $\alpha=0$  drops to 4/5 with a seed at  $\bar{C}=8.46$ . The over-provisioned config ( $\alpha=0.5, h=5$ ) *regresses* to 4/5 ( $\bar{C}=9.26$  outlier) — large- $h$  BRT shaping can integrate spurious penalty (see the  $\Delta t$  sensitivity above) and destabilise training. The DT-margin recipe

Table 3: DT-margin  $\alpha/h$  sensitivity on Goal (5 seeds  $\times$  200K).

| Config              | $\alpha$ | $h$ | mean $\bar{C}$                  | GO/5       |
|---------------------|----------|-----|---------------------------------|------------|
| no margin           | 0.0      | 2   | 3.37 $\pm$ 2.81                 | 4/5        |
| short horizon       | 0.05     | 1   | 1.94 $\pm$ 1.78                 | 4/5        |
| both at minimum     | 0.05     | 2   | 2.36 $\pm$ 2.60                 | 4/5        |
| above minimum       | 0.1      | 2   | <b>1.30<math>\pm</math>1.66</b> | <b>5/5</b> |
| $\alpha$ generous   | 0.3      | 1   | 1.72 $\pm$ 1.46                 | 5/5        |
| <b>paper choice</b> | 0.3      | 3   | <b>0.96<math>\pm</math>0.68</b> | <b>5/5</b> |
| over-provisioned    | 0.5      | 5   | 3.00 $\pm$ 3.38                 | 4/5        |

is thus robust only with margin on both knobs; VA-ATACOM eliminates  $\alpha$  entirely, replacing it with the calibrated  $a_{\max}$ .

### Statistical Significance

**Safety-Gymnasium (Table 1 of the main paper).** We pair every (task, seed) of each baseline against the same (task, seed) of VA-ATACOM (15 paired episode-cost samples per baseline; 11 baselines  $\times$  15 = 165 paired tests in total). One-sided Wilcoxon signed-rank tests (alternative: baseline cost > VA-ATACOM cost) give:

| Baseline                      | $p$ -value           | sig. |
|-------------------------------|----------------------|------|
| PPO baseline                  | $3.1 \times 10^{-5}$ | ††   |
| ATACOM (null)                 | $3.1 \times 10^{-5}$ | ††   |
| ATACOM-VD                     | $3.1 \times 10^{-5}$ | ††   |
| ATACOM-S                      | $3.1 \times 10^{-5}$ | ††   |
| ATACOM-LA                     | $3.1 \times 10^{-5}$ | ††   |
| HOCBF                         | $3.1 \times 10^{-5}$ | ††   |
| DCM                           | $3.1 \times 10^{-5}$ | ††   |
| Predictive ATACOM             | $4.9 \times 10^{-4}$ | ††   |
| CBF-QP                        | $3.1 \times 10^{-5}$ | ††   |
| PPO-Lag                       | $3.1 \times 10^{-5}$ | ††   |
| DT-margin (1st-order approx.) | $2.9 \times 10^{-3}$ | †    |

†:  $p < 0.05$ , ††:  $p < 0.001$

All 11 baselines are significantly worse than VA-ATACOM ( $p < 0.05$ ); 10 are extremely significant ( $p < 0.001$ ). DT-margin, the first-order Taylor approximation of VA-ATACOM, has the smallest gap ( $p=0.003$ ) — consistent with the main paper’s claim that the linearisation captures most of the safe behaviour but leaves a residual gap that only the exact quadratic constraint closes (10/15 paired cells where DT-margin > VA-ATACOM, 4/15 ties at 0). A Friedman omnibus test across all 12 methods rejects equality with  $\chi^2=97.05$ ,  $p=6.8 \times 10^{-16}$  (df=11).

**Webots transfer (120 paired trials).** For the Webots E-puck study (§ Sim-to-Real of main paper) we apply the McNemar test for paired binary outcomes (deep penetration  $\geq 1$  cm yes/no) on the same (start, goal, world, controller) configuration with the filter on vs. off:

The filter *saves* 23 paired trials (filter on safe, filter off deep) and *hurts* 0 paired trials (filter on deep, filter off safe). The number-needed-to-filter is  $N/\text{saves} = 120/23 \approx 5.2$ : on

| Setting                 | N   | on-deep | off-deep | saves | $p$                                    |
|-------------------------|-----|---------|----------|-------|----------------------------------------|
| P-controller (3 worlds) | 60  | 1       | 14       | 13    | $1.2 \times 10^{-4}$                   |
| PPO transfer (3 worlds) | 60  | 0       | 10       | 10    | $9.8 \times 10^{-4}$                   |
| <b>All 120 trials</b>   | 120 | 1       | 24       | 23    | <b><math>1.2 \times 10^{-7}</math></b> |

average one in every five trajectories needs the filter to avoid deep penetration. The one VA-ATACOM deep case (dense, P-controller, triple-obstacle pocket) is *also* deep without the filter, so the filter never makes things worse.

### Generalisation Beyond Point-Robot Dynamics

To probe VA-ATACOM’s scope across different mobile-robot dynamics, we evaluate the unchanged filter on `SafetyCarGoal1-v0`, which replaces the Point agent’s holonomic velocity actions with a bicycle-like throttle+steering. All other env parameters (8 hazards of radius 0.2, BRT frontend, 5 seeds, 200K steps) match the main Table 1.

| Method                | mean $\bar{C}$ | GO/5 |
|-----------------------|----------------|------|
| PPO (no filter)       | 23.20          | 1/5  |
| ATACOM (null-space)   | <b>9.20</b>    | 1/5  |
| DT-margin (heuristic) | 13.02          | 1/5  |
| <b>VA-ATACOM</b>      | 23.92          | 1/5  |

Every method is NO-GO under the 200K budget, but the relative ordering *inverts* compared with the Point-robot Table 1: on Car, the null-space ATACOM (9.20) and DT-margin (13.02) beat VA-ATACOM (23.92).

The cause is structural and identified in our main paper: VA-ATACOM’s tangent-cone projection is currently implemented with the `action_form='diff_drive'` constraint (§ VA-ATACOM Algorithm), which assumes *forward speed and angular rate are independent actuators*. Car’s bicycle model couples them through the wheelbase, so reducing forward command also reduces steerability — the filter’s safety brake removes the very degree of freedom the agent needs to turn away from the obstacle. The null-space ATACOM projection, by contrast, projects onto the Jacobian null-space of the position constraint, which is independent of the actuation parameterisation and degrades gracefully on Car.

This is a clear scope clarification for VA-ATACOM, not a refutation of Prop. 4 itself: the manifold and its discrete-time invariance theorem hold; the bicycle-Car case requires a different action-form projection (matching steering Jacobian) for the implementation. Designing a bicycle-aware action form is an obvious follow-up and is listed in the main paper’s Limitations L4.

### Anticipated Questions

We pre-emptively address questions reviewers commonly raise on safety-filter papers.

**Q1. Webots is not real hardware — why claim "sim-to-real transfer"?** We use the phrase narrowly. Safety-Gymnasium is a 2D point-mass abstraction; Webots is a 3D rigid-body simulator (Cyberbotics) with full ODE contact dynamics, a physical E-puck mesh (54.0 mm diameter, 53.0 mm wheel-base, accurate inertia and motor torque), and *raw noisy GPS*/heading feeding the controller — no state estimator. The transfer claim is therefore: from a holonomic point mass at 10 Hz to a differential-drive rigid body at 16 Hz with sensor noise, *the unchanged filter* reduces deep penetrations  $24 \rightarrow 1$  in 120 paired trials. Physical-E-puck experiments using UWB localisation are listed as on-going work; the protocol is in our public repository.

**Q2. Is VA-ATACOM just ATACOM + a velocity term?** No. (i) The velocity augmentation alone is insufficient (Design row *model-free braking scaling*, cost 117.4, 0/5 GO — highest in Table 2); the tangent-cone projection onto the velocity-augmented manifold is what handles momentum. (ii) Proposition 4 is, to our knowledge, the first *discrete-time* forward-invariance theorem for the ATACOM-style tangent projection, with an explicit chord-defect  $\delta = \frac{3}{2} \Delta t^2 a_{\max}$  and constructive conditions on  $(d_{\text{safe}}, \alpha_0)$  from  $(\Delta t, r, v_{\max})$ . (iii) Both ATACOM and the velocity-adaptive margin are recovered as limits/approximations (Cor. 1,  $\Delta t \rightarrow 0$ ; first-order Taylor in  $\|v\|$ ).

**Q3. DT-margin reaches 13/15 GO; VA-ATACOM 15/15. Is the 2-cell gain meaningful?** Three arguments. (a) DT-margin requires hand-tuning  $\alpha$  and a BRT horizon  $h$  (see DT-margin sensitivity tables above); VA-ATACOM has zero hyperparameters once  $a_{\max}$  is calibrated. (b) The Wilcoxon paired test rejects equality with  $p = 2.9 \times 10^{-3}$ , and the 4 pairs that tie are exactly where both methods reach  $\bar{C} = 0$  — in non-trivial cells DT-margin is strictly worse 10/11 times. (c) Theoretically, DT-margin is the first-order Taylor of VA-ATACOM in  $\|v\|$ , so the residual gap is structural, not noise.

**Q4. Are the Prop. 4 assumptions  $(\alpha_0 \Delta t \leq 1, d_{\text{safe}} \geq \frac{3}{2} \Delta t a_{\max} / \alpha_0, r \geq d_{\text{safe}} + \Delta t v_{\max})$  restrictive?** They are constructive, not restrictive. At Safety-Gym defaults ( $\Delta t = 0.1$  s,  $v_{\max} = 1$  m/s,  $a_{\max} = 0.9$  m/s<sup>2</sup>,  $\alpha_0 = 1$ ), the bounds give  $d_{\text{safe}} \geq 0.135$  m and  $r \geq 0.235$  m, both met by the env defaults ( $r = 0.2/0.3$ ). The first two come from the discretisation defect  $\delta$  in Step 2 of the proof; the third is a steady-state distance budget. All three are checked at run-time and printed in our logs.

**Q5. Why no GPU? Would a larger PPO net change the picture?** The bottleneck is environment simulation, not the policy (60-dim obs, 64-64 MLP). On a 28-thread CPU we run 20 parallel 200K trainings in 10 h wall-clock; a GPU yields a small speed-up but the failure modes (M1/M2) are about the constraint, not the model class. We trained PPO-Lag (the only filter-free safe-RL method in our sweep) on the same 64-64 MLP for fair comparison; it reaches 1/15 GO.

- **CPU:** Intel Core i7-13850HX (20 cores, 28 threads).
- **RAM:** 31 GiB DDR5.
- **OS:** Ubuntu 24.04 LTS, Linux kernel 6.17.
- **GPU:** not used — all training and evaluation run *single-threaded on CPU* (OMP\_NUM\_THREADS=1, MKL\_NUM\_THREADS=1). Each Safety-Gymnasium 200K-step run takes  $\approx 15$ –20 min wall-clock; we parallelise up to 20 such single-threaded runs at once.

Software stack (versions used at experiment time):

- Python 3.10
- PyTorch 2.x (CPU build)
- NumPy 1.23 / matplotlib 3.x
- safety-gymnasium 1.0 (the Safety-Gym successor; we ran with SafetyPointGoal1-v0, SafetyPointPush1-v0, SafetyPointMultiGoal0-v0)
- gymnasium 0.28 / mujoco 2.3
- Webots R2023b (for the rigid-body validation in main § Sim-to-Real)

Random seeds 0, 1, 2, 3, 4 are used for every (method, task) cell in Table 1 of the main paper. Total compute for the full Table 1 + Webots study is approximately **200 CPU-hours** on the workstation above ( $\sim 10$  h wall-clock with 20-way parallelism); the L-shape audit re-run added a further  $\sim 5$  CPU-hours.

## Computing Infrastructure

All experiments reported in the main paper and this supplementary were run on a single workstation: